Quantitative Methods for Management

Tutorial Exercise Sheet: Complex Numbers and Matrix Powers

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Recap: Representations of complex numbers. A nonzero complex number $z \in \mathbb{C}$ can be written in two different forms:

$$\begin{aligned} \text{Cartesian form:} \quad & z = a + bi, \\ \text{Polar form:} \quad & z = r(\cos\theta + i\sin\theta), \end{aligned}$$

where $a, b \in \mathbb{R}$, the radius $r = |z| = \sqrt{a^2 + b^2} \geq 0$ is the distance from the origin, and θ is the angle measured counterclockwise from the positive real axis, in radians. The relations

$$a = r\cos\theta, \qquad b = r\sin\theta$$

help determine the angle. Note that angles that differ by an integer multiple of 2π represent the same direction. Thus, it is sufficient to choose $\theta \in [0, 2\pi)$. The number 0 has radius 0 and no defined angle.

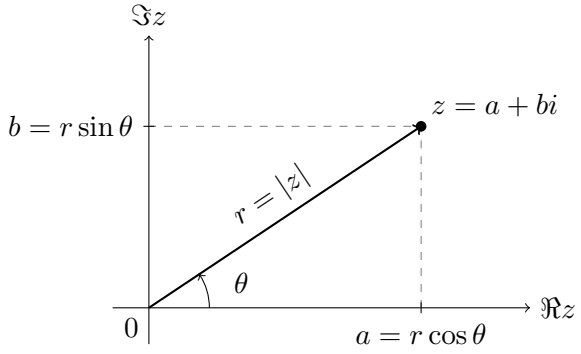


Figure 1: Different representations of a complex number in the complex plane.

Exercise 1 (Complex numbers and basic arithmetic)**.** Simplify each expression to the form $a + bi$ and plot the resulting number in the complex plane. Recall that $i^2 = -1$.

(a) $(-3 + 3i) + (1 - 5i)$

(b) $(2 + i) - (4 - 3i)$

(c) $(1 + 2i)(1 + 3i)$

(d) $\dfrac{3 + i}{1 - i}$

(e) $\dfrac{2 + 4i}{2i}$

Solution. For $z = a+bi$, the real part is a and the imaginary part is b. The corresponding point in the complex plane is (a, b). To divide by a complex number, multiply the numerator and denominator by the complex conjugate of the denominator.

(a) Combine real and imaginary parts:

$$(-3 + 3i) + (1 - 5i) = (-3 + 1) + (3 - 5)i = -2 - 2i.$$

(b) Distribute the minus sign and combine terms:

$$(2 + i) - (4 - 3i) = 2 + i - 4 + 3i = -2 + 4i.$$

(c) Expand the product and use $i^2 = -1$:

$$(1 + 2i)(1 + 3i) = 1 + 3i + 2i + 6i^2 = -5 + 5i.$$

(d) Multiply by the conjugate $1 + i$:

$$\frac{3 + i}{1 - i} = \frac{(3 + i)(1 + i)}{(1 - i)(1 + i)} = \frac{2 + 4i}{2} = 1 + 2i.$$

(e) Use $1/i = -i$:

$$\frac{2 + 4i}{2i} = \frac{1}{i} + 2 = 2 - i.$$

Complex numbers in the complex plane

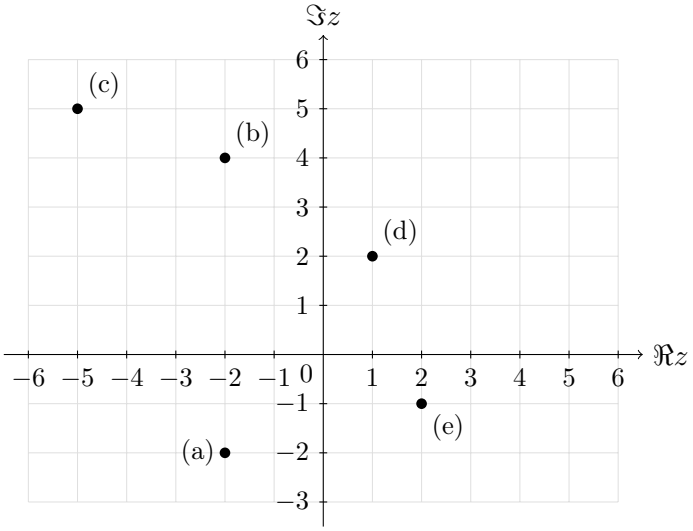


Figure 2: The plot of Exercise 1.

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Exercise 2 (Polar representations)**.** Find the radius and angle of each complex number.

(a) $1 + i$

(b) $-1 + i$

(c) $-2i$

Solution. (a) For $1+i$, the radius is $r=\sqrt{1^2+1^2}=\sqrt{2}$. The point $(1,1)$ is an $\frac{1}{8}$th rotation, $\theta = 2\pi/8 = \pi/4$. Hence,

$$1+i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right).$$

(b) For $-1+i$, the radius is $r=\sqrt{(-1)^2+1^2}=\sqrt{2}$. The point $(-1,1)$ is an $\frac{3}{8}$th rotation, so $\theta = 6\pi/8 = 3\pi/4$. Hence,

$$-1+i = \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right).$$

(c) For $-2i$, the radius is $r=\sqrt{0^2+(-2)^2}=2$. The point $(0,-2)$ is an $\frac{3}{4}$th rotation, so $\theta = 6\pi/4 = 3\pi/2$. Hence,

$$-2i = 2\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right).$$

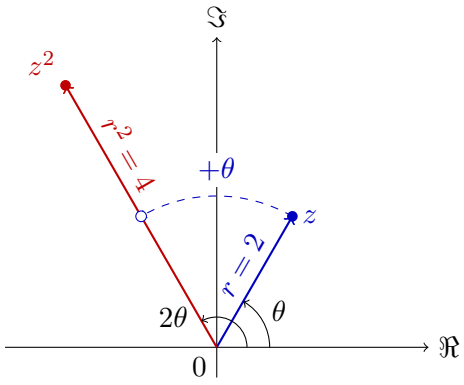
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Recap: De Moivre's formula. For $z = r(\cos\theta + i\sin\theta)$ and a positive integer n,

$$\big[r(\cos\theta + i\sin\theta)\big]^n = r^n\big(\cos(n\theta) + i\sin(n\theta)\big).$$

Multiplying a complex number by z rotates its position around the origin by the angle θ and multiplies its radius by r. Starting at 1 on the real axis, multiplying by z repeatedly gives $z, z^2, z^3, \ldots$ Each step adds θ to the angle and multiplies the radius by r. After n steps, the angle is $n\theta$ and the radius is r^n. Full turns of 2π do not change the final direction. If $r=1$, each step is a rotation only; if $r>1$, the radius increases, and if $0<r<1$, it decreases.

One multiplication step: from z to z^2



Exercise 3 (De Moivre's formula)**.** Use De Moivre's formula to calculate the following powers. Express each result in the form $a+bi$.

(a) $(1+i)^4$

(b) $(1-i)^6$

(c) $(-1+i)^3$

Solution. Write each number in polar form, raise its radius to the given power, and multiply its angle by that power.

(a) The number $1+i$ has radius $\sqrt{2}$ and angle $\pi/4$. Therefore,

$$\begin{aligned}(1+i)^4 &= (\sqrt{2})^4\,(\cos\pi + i\sin\pi)\\ &= 4(-1+0i) = -4.\end{aligned}$$

(b) The number $1-i$ has radius $\sqrt{2}$ and angle $7\pi/4$. Multiplying the angle by 6 gives $21\pi/2 = \pi/2 + 5\cdot 2\pi$. Hence,

$$\begin{aligned}(1-i)^6 &= (\sqrt{2})^6\left(\cos\frac{21\pi}{2} + i\sin\frac{21\pi}{2}\right)\\ &= 8\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 8i.\end{aligned}$$

(c) The number $-1+i$ has radius $\sqrt{2}$ and angle $3\pi/4$. Multiplying the angle by 3 gives $9\pi/4 = \pi/4 + 2\pi$. Hence,

$$\begin{aligned}(-1+i)^3 &= (\sqrt{2})^3\left(\cos\frac{9\pi}{4} + i\sin\frac{9\pi}{4}\right)\\ &= 2\sqrt{2}\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = 2+2i.\end{aligned}$$

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Recap: Powers of diagonalizable matrices. For a square matrix $\mathbf{A}$, the power $\mathbf{A}^n$ means multiplying $\mathbf{A}$ by itself n times, rather than raising each entry to the power n. Eigenvalues and eigenvectors provide a convenient way to perform this calculation. If a 2×2 matrix $\mathbf{A}$ has two linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2$, with corresponding eigenvalues λ_1, λ_2, we form

$$\mathbf{P} = \begin{pmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

The eigenvalues in $\mathbf{D}$ must follow the same order as the eigenvectors in $\mathbf{P}$. Since $\mathbf{A}\mathbf{v}_1 = \lambda_1\mathbf{v}_1$ and $\mathbf{A}\mathbf{v}_2 = \lambda_2\mathbf{v}_2$, multiplying $\mathbf{A}$ by the columns of $\mathbf{P}$ gives

$$\mathbf{AP} = \begin{pmatrix} | & | \\ \lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 \\ | & | \end{pmatrix} = \mathbf{PD}.$$

The columns of $\mathbf{P}$ are linearly independent, so $\mathbf{P}$ is invertible. Multiplying both sides on the right by $\mathbf{P}^{-1}$ gives

$$\mathbf{A} = \mathbf{PDP}^{-1}.$$

To derive a formula for the powers of $\mathbf{A}$, we substitute the expression into repeated matrix multiplication. Since $\mathbf{P}^{-1}\mathbf{P} = I$ with I the identity matrix, we obtain for $n \geq 2$

$$\begin{aligned}\mathbf{A}^n &= \underbrace{(\mathbf{PDP}^{-1}) \cdots (\mathbf{PDP}^{-1})}_{n \text{ factors}} \\ &= \mathbf{PD}\underbrace{(\mathbf{P}^{-1}\mathbf{P})}_{I}\mathbf{D} \cdots \underbrace{(\mathbf{P}^{-1}\mathbf{P})}_{I}\mathbf{DP}^{-1} \\ &= \mathbf{P}\underbrace{\mathbf{D} \cdots \mathbf{D}}_{n \text{ factors}}\mathbf{P}^{-1} = \mathbf{PD}^n\mathbf{P}^{-1}.\end{aligned}$$

We only regroup the factors. Their order stays the same. The $n-1$ intervening pairs $\mathbf{P}^{-1}\mathbf{P}$ equal I, leaving n factors $\mathbf{D}$ between $\mathbf{P}$ and $\mathbf{P}^{-1}$. For $n = 1$, the formula is the diagonalization itself. Thus, for every positive integer n,

$$\mathbf{A}^n = \mathbf{PD}^n\mathbf{P}^{-1}, \qquad \mathbf{D}^n = \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix}.$$

Because $\mathbf{D}$ is diagonal, multiplying it by itself simply multiplies the corresponding diagonal entries. We therefore need only the scalar powers λ_1^n and λ_2^n, followed by multiplication with $\mathbf{P}$ and $\mathbf{P}^{-1}$. Also, $\mathbf{A}^0 = I$. The eigenvalues and eigenvectors may be complex, even when the entries of $\mathbf{A}$ are real.

- **Find the eigenvalues.** Solve $\det(\mathbf{A} - \lambda I) = 0$ and place the eigenvalues on the diagonal of $\mathbf{D}$.
- **Find the eigenvectors.** Solve $(\mathbf{A} - \lambda I)\mathbf{v} = 0$ for each eigenvalue. Choose two linearly independent eigenvectors and use them as the columns of $\mathbf{P}$, in the corresponding order. If this is impossible, this method does not apply.
- **Calculate the power.** Compute $\mathbf{P}^{-1}$, raise each diagonal entry of $\mathbf{D}$ to n, and calculate $\mathbf{PD}^n\mathbf{P}^{-1}$.

Along an eigenvector, multiplication by $\mathbf{A}$ scales the vector by its eigenvalue. Repeating this n times scales it by the nth power of that eigenvalue.

Exercise 4 (Matrix powers with real eigenvalues)**.** Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

(a) Find the eigenvalues of $\mathbf{A}$ and form $\mathbf{D}$.

(b) Find corresponding eigenvectors, form $\mathbf{P}$, and calculate $\mathbf{P}^{-1}$.

(c) Use the diagonal form to calculate $\mathbf{A}^3$.

(d) Find a formula for $\mathbf{A}^n$, for positive integers n.

Solution. (a) First, we compute

$$\det(\mathbf{A} - \lambda I) = (2-\lambda)^2 - 1 = (\lambda - 3)(\lambda - 1).$$

The eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 1$, so we choose

$$\mathbf{D} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

(b) Write $\mathbf{v} = (v_1, v_2)^T$. For $\lambda_1 = 3$, the equation $(\mathbf{A} - 3I)\mathbf{v} = 0$ gives $-v_1 + v_2 = 0$, so $v_2 = v_1$. For $\lambda_2 = 1$, the equation $(\mathbf{A} - I)\mathbf{v} = 0$ gives $v_1 + v_2 = 0$, so $v_2 = -v_1$. Choosing $v_1 = 1$ gives

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

These eigenvectors are linearly independent. In the same order as the eigenvalues in $\mathbf{D}$, we obtain

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det(\mathbf{P}) = 1 \cdot (-1) - 1 \cdot 1 = -2.$$

Hence,

$$\mathbf{P}^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

(c) We raise the diagonal entries of $\mathbf{D}$ to the third power and then multiply:

$$\begin{aligned} \mathbf{A}^3 &= \mathbf{P} \begin{pmatrix} 27 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{P}^{-1} \\ &= \frac{1}{2} \begin{pmatrix} 28 & 26 \\ 26 & 28 \end{pmatrix} = \begin{pmatrix} 14 & 13 \\ 13 & 14 \end{pmatrix}. \end{aligned}$$

(d) For any positive integer n, the same calculation gives

$$\begin{aligned} \mathbf{A}^n &= \mathbf{P} \begin{pmatrix} 3^n & 0 \\ 0 & 1 \end{pmatrix} \mathbf{P}^{-1} \\ &= \frac{1}{2} \begin{pmatrix} 3^n + 1 & 3^n - 1 \\ 3^n - 1 & 3^n + 1 \end{pmatrix}. \end{aligned}$$

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Exercise 5 (Matrix powers with complex eigenvalues)**.** Let

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

(a) Find the eigenvalues of $\mathbf{A}$ and form $\mathbf{D}$.

(b) Find corresponding eigenvectors, form $\mathbf{P}$, and calculate $\mathbf{P}^{-1}$.

(c) Use De Moivre's formula to calculate the diagonal entries of $\mathbf{D}^4$, then compute $\mathbf{A}^4 = \mathbf{P}\mathbf{D}^4\mathbf{P}^{-1}$.

(d) Find a formula for $\mathbf{A}^n$, for positive integers n, using the polar representation.

Solution. (a) First, we compute

$$\det(\mathbf{A} - \lambda I) = (1 - \lambda)^2 + 1.$$

Setting this expression equal to zero gives $(1 - \lambda)^2 = -1$. The eigenvalues are $\lambda_1 = 1 + i$ and $\lambda_2 = 1 - i$, so we choose

$$\mathbf{D} = \begin{pmatrix} 1 + i & 0 \\ 0 & 1 - i \end{pmatrix}.$$

(b) Write $\mathbf{v} = (v_1, v_2)^T$. For $\lambda_1 = 1 + i$, the equation $(\mathbf{A} - (1+i)I)\mathbf{v} = 0$ gives $-iv_1 - v_2 = 0$, so $v_2 = -iv_1$. For $\lambda_2 = 1 - i$, the equation $(\mathbf{A} - (1-i)I)\mathbf{v} = 0$ gives $iv_1 - v_2 = 0$, so $v_2 = iv_1$. Choosing $v_1 = 1$ gives

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

These eigenvectors are linearly independent. In the same order as the eigenvalues in $\mathbf{D}$, we obtain

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}, \qquad \det(\mathbf{P}) = 1 \cdot i - 1 \cdot (-i) = 2i.$$

Hence,

$$\mathbf{P}^{-1} = \frac{1}{2i} \begin{pmatrix} i & -1 \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}.$$

(c) Both eigenvalues have radius $\sqrt{2}$. Their angles are $\pi/4$ and $7\pi/4$, respectively. De Moivre's formula gives

$$(1+i)^4 = (\sqrt{2})^4 (\cos\pi + i\sin\pi) = -4,$$
$$(1-i)^4 = (\sqrt{2})^4 (\cos(7\pi) + i\sin(7\pi)) = -4.$$

We then compute

$$\mathbf{A}^4 = \mathbf{P} \begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix} \mathbf{P}^{-1}$$
$$= -4\mathbf{P}\mathbf{P}^{-1} = -4I = \begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix}.$$

(d) Since $7\pi/4$ and $-\pi/4$ describe the same direction, De Moivre's formula gives

$$(1 \pm i)^n = 2^{n/2} \left(\cos\frac{n\pi}{4} \pm i \sin\frac{n\pi}{4} \right).$$

Write $c_n = \cos(n\pi/4)$ and $s_n = \sin(n\pi/4)$. Substituting these powers into $\mathbf{D}^n$ and multiplying gives

$$\mathbf{A}^n = \frac{2^{n/2}}{2} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \begin{pmatrix} c_n + is_n & 0 \\ 0 & c_n - is_n \end{pmatrix} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$$
$$= 2^{n/2} \begin{pmatrix} c_n & -s_n \\ s_n & c_n \end{pmatrix} = 2^{n/2} \begin{pmatrix} \cos(n\pi/4) & -\sin(n\pi/4) \\ \sin(n\pi/4) & \cos(n\pi/4) \end{pmatrix}.$$

The resulting matrix has real entries, as expected because $\mathbf{A}$ has real entries.

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Recap: graphs and transition matrices. A directed graph consists of vertices and arrows connecting them. Its adjacency matrix $\mathbf{A}$ records these connections. We use the convention

$$A_{ij} = \begin{cases} 1, & \text{if there is an arrow from vertex } j \text{ to vertex } i, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, column j describes the arrows leaving vertex j, and

$$d_j = \sum_i A_{ij}$$

counts its outgoing arrows.

Suppose a traveler moves along one outgoing arrow at each step, choosing each available arrow with equal probability. The next move depends only on the current vertex. We obtain the Markov transition matrix $\mathbf{P}$ by dividing each column by its sum:

$$P_{ij} = \frac{A_{ij}}{d_j}, \qquad d_j > 0.$$

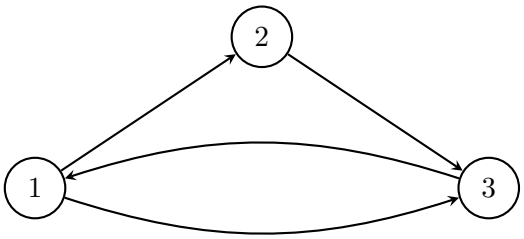
Here, P_{ij} is the probability of moving from j to i in one step. All entries are nonnegative, and every column sums to 1.

- **Build the adjacency matrix.** Choose an order for the vertices and enter each arrow in its destination row and starting column.
- **Normalize the columns.** Divide each column by the number of outgoing arrows. If a vertex has no outgoing arrows, we let the traveler stay there: set its diagonal entry in $\mathbf{P}$ to 1 and all other entries in that column to 0.
- **Update the probabilities.** Let $\mathbf{p}^{(k)}$ be the column vector containing the probabilities of being at each vertex after k steps. Its entries are nonnegative and sum to 1. Then

  $$\mathbf{p}^{(k+1)} = \mathbf{P}\mathbf{p}^{(k)}, \qquad \mathbf{p}^{(n)} = \mathbf{P}^n\mathbf{p}^{(0)}.$$

  The entry $(\mathbf{P}^n)_{ij}$ gives the probability of reaching i after n steps when starting at j.

Exercise 6 (A random walk on a graph)**.** Consider the directed graph below. At each step, a traveler chooses uniformly among the outgoing arrows.



(a) Construct the adjacency matrix $\mathbf{A}$ using the vertex order $1, 2, 3$. Use columns for starting vertices.

(b) Determine the number of outgoing arrows at each vertex. Construct the transition matrix $\mathbf{P}$ and verify that each column sums to 1.

(c) Starting at vertex 1, calculate the probability distributions after one and two steps.

(d) Calculate $\mathbf{P}^2$. Interpret its entry in row 3, column 1, using the possible two-step paths.

Solution. (a) The arrows are $1 \to 2$, $1 \to 3$, $2 \to 3$, and $3 \to 1$. With destination vertices in the rows and starting vertices in the columns, we obtain

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

(b) The column sums are $d_1 = 2$, $d_2 = 1$, and $d_3 = 1$. We divide the first column by 2 and the other columns by 1:

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & 0 \end{pmatrix}.$$

All entries are nonnegative. The column sums are $0 + \frac{1}{2} + \frac{1}{2} = 1$, $0 + 0 + 1 = 1$, and $1 + 0 + 0 = 1$, as required.

(c) Since the traveler starts at vertex 1, the initial distribution is $\mathbf{p}^{(0)} = (1, 0, 0)^T$. We compute

$$\mathbf{p}^{(1)} = \mathbf{P} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix},$$

$$\mathbf{p}^{(2)} = \mathbf{P} \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix}.$$

After one step, the traveler is at vertex 2 or 3, each with probability $1/2$. After two steps, the traveler is at vertex 1 or 3, each with probability $1/2$.

(d) Matrix multiplication gives

$$\mathbf{P}^2 = \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}.$$

The entry $(\mathbf{P}^2)_{31} = 1/2$ is the probability of reaching vertex 3 after two steps when starting at vertex 1. The only two-step paths from vertex 1 are $1 \to 2 \to 3$ and $1 \to 3 \to 1$. Only the first ends at vertex 3, and its probability is $\frac{1}{2} \cdot 1 = \frac{1}{2}$.

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